

Lecture 01 — Overview of Metabolic Pathways

Complete study guide — PowerPoint (all 41 slides + speaker notes) and lecture transcript, reconciled

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SESSION Tuesday 8/4, 1:00–2:00 PM, in person (Poll Everywhere participation)

REQUIRED READING Lippincott Illustrated Reviews: Biochemistry, Ch. 8 (Introduction to Metabolism) · **SUGGESTED** Marks' Basic Medical Biochemistry 5th ed., Ch. 1 and Ch. 9

ASSESSED ON Foundation Exam #1, 8/17, 11 AM–1 PM (with Enzymes & Kinetics, ETC, Thermodynamics primer, Signal Transduction, Carbohydrate digestion, Carbohydrate structures)

HOW TO READ THIS DOCUMENT

Blue = the concept, the why. **Gold** = the specifics to memorize. **Red** = traps and "always/never" statements Dr. Bagh repeated. **Green** = clinical. **Grey** = corrections where the slides or the spoken lecture were imprecise. Each fact appears once, in whichever block it belongs to — the blue blocks explain, the gold blocks list, and they do not restate each other.

What this lecture actually is: scaffolding, not content. Dr. Bagh said some version of "you will learn this later" more than twenty times. There is not a single pathway here you must be able to draw. What you must own is the vocabulary and the rules every later pathway will obey — catabolism vs anabolism, which carrier does which job, what makes a step irreversible, what "committed" and "rate-limiting" mean, and the ways a cell turns an enzyme up or down.

Part 0 / The professor's own mental map

Dr. Bagh opened by putting her own handwritten summary on screen before any slide content — "that's how I think the lecture has to be introduced." It is the spine of everything that follows. (The scanned image on slide 4 is cut off at the bottom; the missing lines, which she spoke, are restored here.)

METABOLISM

— CATABOLISM

| **Macronutrients** (carbohydrate, lipid, protein) + **Micronutrients** (vitamins, minerals)

| ↓

| **OXIDATION** (in the body = **loss of hydrogen**, not gain of oxygen)

| • Glycolysis →→ acetyl-CoA • TCA cycle → CO₂ + H₂O • HMP shunt → NADPH + pentoses (no ATP)

| **Burn carbohydrate** (↑ insulin) →→ acetyl-CoA + ATP **Burn fat** (↓ insulin) →→ acetyl-CoA + ATP

| **NADH → ATP** **NADPH × ATP**

| — **ANABOLISM** CH₃—CH₂—CH₂—COOH (e.g. a fatty acid — needs C and H)

• **ATP** — the universal requirement • **Acetyl-CoA** — source of carbon • **NADPH** — source of hydrogen

• ↑ **Insulin** — the anabolic hormone ... **gluconeogenesis** **ketogenesis**

THE WHOLE LECTURE IN ONE PARAGRAPH

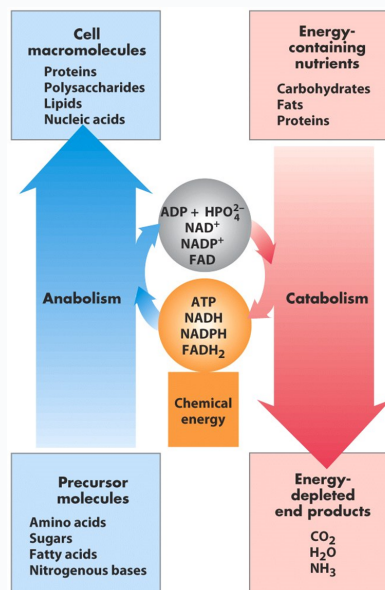
You eat macronutrients. They can't be used as eaten, so they are digested to their simplest absorbable forms, absorbed, and taken up by cells. Inside the cell, **catabolism** oxidizes them — and oxidation here means *stripping hydrogens off*, not adding oxygen. Those hydrogens must land somewhere, and the things that catch them are NAD⁺, NADP⁺ and FAD, which become NADH, NADPH and FADH₂. NADH and FADH₂ are cashed in for ATP. Once ATP is plentiful — a **high energy charge** — the cell flips to **anabolism**. Building needs energy (ATP), carbon (acetyl-CoA) and hydrogen (NADPH), all three of which came out of catabolism. The hormone that turns the anabolic program on is insulin.

Objective 1 / Compare and contrast catabolic and anabolic pathways

WHAT METABOLISM IS

Metabolism is the sum of all chemical changes in a cell, tissue, or body. Reactions almost never happen in isolation — they are organized into multi-step **pathways**, and pathways intersect into one integrated network. Every pathway is either **degradative (catabolic)** or **synthetic (anabolic)**.

The Stanford "Pathways of Human Metabolism" map on slide 3 (also in the repo as FullSubwayMap222.pdf) exists to make one point: carbohydrate, lipid and protein metabolism are not three separate subjects — they share intermediates and feed each other. You are not expected to memorize it, only to keep returning to it as each pathway is taught.



Slide 6 — the central wheel. Two counter-rotating arrows sharing one hub: catabolism converts the **oxidized** carriers ($\text{ADP} + \text{HPO}_4^{2-}$, NAD^+ , NADP^+ , FAD) into the **charged** ones (ATP , NADH , NADPH , FADH_2), and anabolism spends them.

1.1 Catabolism

THE FIVE DEFINING FEATURES (SLIDE 7, VERBATIM)

1. **Breakdown** of complex molecules into simple ones
2. Generally **exergonic** — proceeds with release of free energy
3. **Releases energy**, captured as ATP
4. **Oxidative** — $\text{NAD}^+ / \text{NADP}^+ \rightarrow \text{NADH} / \text{NADPH}$ (and $\text{FAD} \rightarrow \text{FADH}_2$)
5. **Provides the energy for the synthesis** of complex molecules — it exists to fund anabolism

HER MOST-REPEATED POINT IN THE ENTIRE LECTURE

Oxidation in the human body is by LOSS OF HYDROGEN, not gain of oxygen. She said this at least eight times. Biological oxidation is *dehydrogenation* — hence the enzymes are called **dehydrogenases**. Her follow-on question, asked and answered repeatedly: "and who accepts those hydrogens?" The **acceptors of reducing equivalents** — NAD^+ , NADP^+ , FAD . She defined a reducing equivalent as $2\text{H}^+ + 2\text{e}^-$.

PRECISION ON WHAT EACH ACCEPTOR ACTUALLY TAKES

Strictly, a "reducing equivalent" is one electron (or one hydrogen atom); the $2\text{H}^+ + 2\text{e}^-$ she describes is what a dehydrogenase removes from the substrate, so her usage is right in context. The two acceptors then differ:

- **NAD⁺** accepts a **hydride** ($\text{H}^- = 1\text{H}^+ + 2\text{e}^-$); the other proton is released into solution — which is why it is always written **NADH + H⁺**, a detail she was careful about.
- **FAD** accepts **both** hydrogens ($2\text{H}^+ + 2\text{e}^- \rightarrow \text{FADH}_2$), releasing no free H^+ .

The three oxidative (catabolic) pathways

She listed this triad roughly six separate times. If one fact from Objective 1 is tested, it is likely this.

Pathway	What it mainly oxidizes	What it produces	Source of ATP?
Glycolysis	Mainly carbohydrate	Pyruvate → acetyl-CoA; NADH; some ATP	Yes
TCA cycle = citric acid cycle = tricarboxylic acid cycle — three names, one cycle	Common final pathway for carbohydrate, lipid <i>and</i> protein; all three converge on acetyl-CoA, the cycle's starting molecule	NADH + H ⁺ , FADH ₂ , CO ₂	Yes (indirectly, via the carriers)
HMP shunt = hexose monophosphate shunt = pentose phosphate pathway	Glucose-6-phosphate — an alternate oxidative route	NADPH and pentoses	NO

TRAP

The HMP shunt is an oxidative pathway that **does not give you ATP**. It makes NADPH (reducing power for synthesis) and pentoses (ribose for nucleotides). Don't let "oxidative pathway" reflexively produce the answer "ATP."

THREE STAGES OF CATABOLISM, AND WHAT GETS ABSORBED

She described the staging verbally — "first they are digested and absorbed as simple molecules, picked up by the cells, and *then* starts catabolism." It is Figure 8.3 of the required Lippincott chapter:

1. **Hydrolysis of complex molecules → building blocks.** Proteins → amino acids; polysaccharides → monosaccharides; triacylglycerols → free fatty acids + glycerol. This is digestion and absorption, before the cell is involved.
2. **Building blocks → acetyl-CoA** and a few simple intermediates. Some ATP captured, but little.
3. **Oxidation of acetyl-CoA** in the TCA cycle, electrons flowing from NADH and FADH₂ to oxygen via **oxidative phosphorylation**. This is where the large ATP yield comes from.

Macronutrient	Major absorbable form	Role she assigned it
Carbohydrate	Glucose (monosaccharide)	Preferred fuel
Fat	Fatty acids	Concentrated fuel
Protein	Amino acids	Mainly building blocks — but also used for energy

Other absorbable forms exist for each and are taught later. Two extras she stated: **alcohol is not "empty calories"** — it supplies real calories, entering metabolism directly at acetyl-CoA. And both carbohydrate and fat yield acetyl-CoA → ATP, but **carbohydrate is burned in the fed, high-insulin state and fat in the low-insulin state**.

1.2 The NADH-versus-NADPH rule

WHY TWO NEARLY IDENTICAL MOLECULES EXIST

NADH and NADPH differ by exactly one phosphate group (on the 2'-OH of the adenosine ribose). Chemically they do the same thing — carry two electrons. That phosphate is a **molecular tag** letting the cell run two separate pools with two separate jobs, and letting enzymes recognize which one they are meant to use. **NAD⁺/NADH** is kept mostly **oxidized**, ready to *accept* electrons from catabolism and pass them to the electron transport chain — catabolic currency. **NADP⁺/NADPH** is kept mostly **reduced**, ready to *donate* electrons into biosynthesis — anabolic currency.

Reductive synthesis = any biosynthetic reaction requiring hydrogen (electrons added to the substrate); the donor is NADPH. Her two examples were **fatty acid synthesis** and **cholesterol synthesis** — look at either molecule and it is nothing but carbon and hydrogen, so the cell must supply both a carbon source (acetyl-CoA) and a hydrogen source (NADPH). The main producer of NADPH in this lecture is the HMP shunt.

NADH and FADH₂ → sources of ATP · NADPH → NEVER a source of ATP

"KINDLY REMEMBER THIS VERY IMPORTANT FACT"

- **NADH** is an **indirect** source of ATP — it is oxidized in the electron transport chain and the resulting proton gradient drives ATP synthase. "Indirect" is her word and it is the correct word.
- **FADH₂** is also a source of ATP by the same route, but **less is produced**, because **NAD⁺** is a far more common acceptor than FAD. That is why she always names NADH first.
- **NADPH is never a source of ATP**. It is the **source of hydrogen** for reductive synthesis.

1.3 Anabolism

SLIDE 9 — THE REQUIREMENTS

- **Synthesis** of complex molecules; generally **endergonic**
- **Energy** as **ATP / GTP** to drive enzyme-catalyzed reactions — e.g. ligases, as in glutamine formation by glutamate-ammonia ligase; GTP as with PEP carboxykinase. Some syntheses use ATP, some GTP; requiring one of them is universal.
- **Reducing power** as **NADPH** (and NADH) to supply electrons for reduction of substrates
- **Carbon skeletons** — e.g. for cholesterol biosynthesis, **acetyl-CoA** is required

Anabolism needs: ① ATP (universal) ② Acetyl-CoA (carbon) ③ NADPH (hydrogen) ④ ↑ Insulin

She paused twice and asked the class to recite that list from memory. Expect to be asked for it.

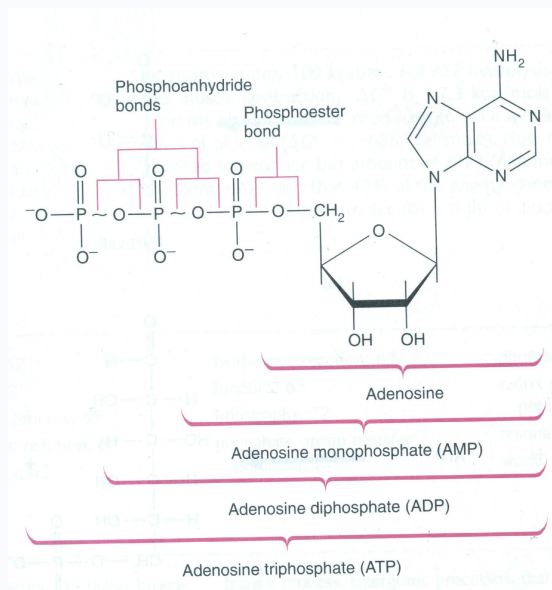
TWO "NEVER PICK THIS" WARNINGS, IN HER WORDS

- **"Please never pick low energy state as active for anabolism."** Anabolism is *ATP-utilizing*; it can only switch on when ATP is already high inside the cell. Low energy state → catabolism.
- **Insulin supports every "-genesis" EXCEPT gluconeogenesis and ketogenesis.** Repeated five times.

WHY INSULIN IS EXEMPT FROM THOSE TWO

Gluconeogenesis (glucose from non-carbohydrate sources) and ketogenesis (ketone bodies) are *synthetic in form* but belong to the **fasting** program — their purpose is to supply fuel when food is absent. Insulin signals "fed," so it actively **inhibits** both. This is the one place where "anabolic = insulin" breaks, which is exactly why it is examinable.

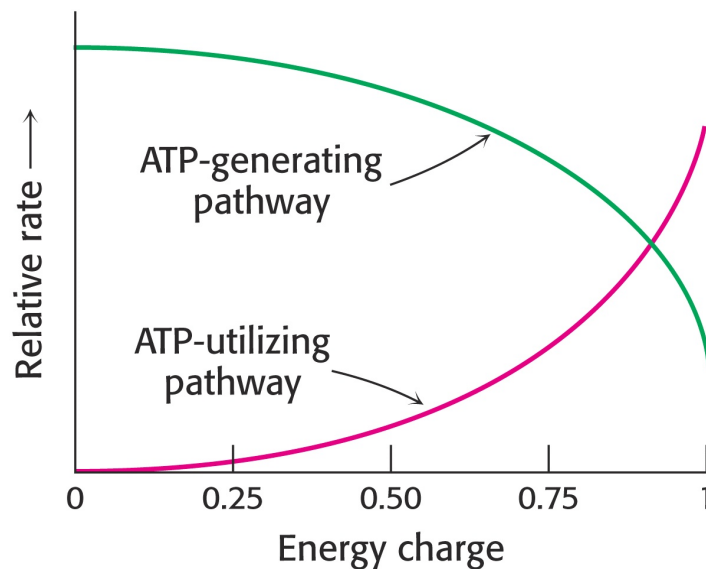
1.4 ATP and the energy charge



Slide 10 — which bonds are the "high-energy" ones. The two bonds *between* phosphates are **phosphoanhydride** bonds (high energy); the bond joining the α -phosphate to ribose is a **phosphoester** bond and is *not*.

ATP FACTS

- ATP is a **high-energy phosphate compound** and the most commonly used energy source in the body; it comes from **catabolism of macronutrients**. A **high-energy compound** is one liberating energy equal to or greater than ATP (others come later in the course).
- "High-energy" bonds are written with a **squiggle** (~).
- **ATP contains TWO high-energy bonds; ADP contains ONE** (the β and γ phosphoanhydride bonds).
- Standard free energy of hydrolysis = $-7,300 \text{ cal/mol}$ (-7.3 kcal/mol) for **both** the β and γ phosphate.
- Definition to be able to state: a **high-energy bond** is a covalent bond whose hydrolysis releases a **useful amount of energy to perform work**.
- **Energy charge** = $([\text{ATP}] + \frac{1}{2} [\text{ADP}]) / ([\text{ATP}] + [\text{ADP}] + [\text{AMP}])$ — ADP counts as half because it carries one of the two high-energy bonds. Range **0** (all AMP) to **1** (all ATP); **buffered like the pH of a cell**, sitting at **0.8–0.95** in most cells.
- **High energy charge** → catabolic pathways **INHIBITED**, anabolic pathways **STIMULATED**.

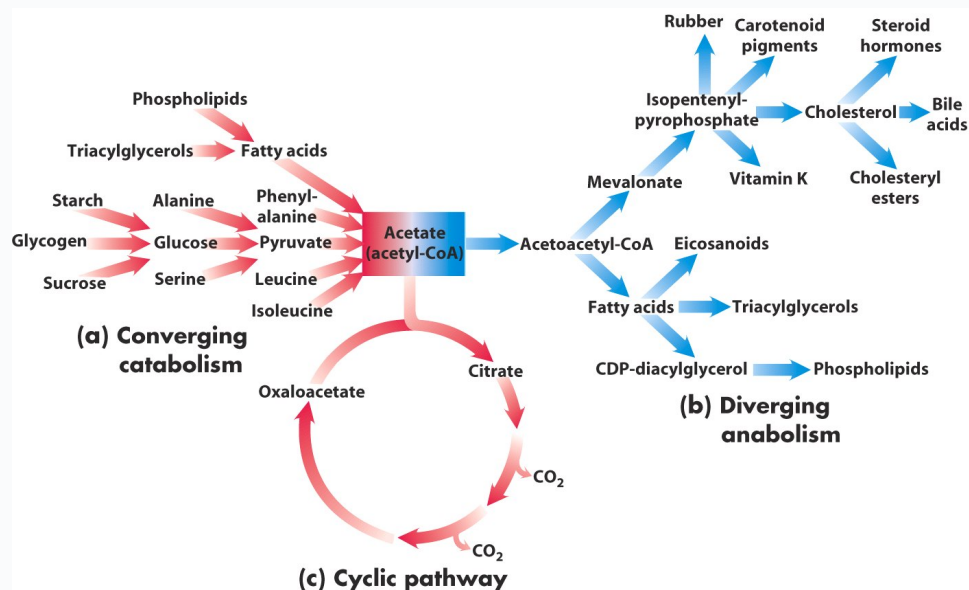


Slide 11 — why the system is stable. The two curves cross at a high energy charge (~0.85–0.9), which is precisely where cells sit, so any drift in either direction is immediately opposed. (Berg/Tymoczko/Stryer 5th ed.)

A MISCONCEPTION WORTH PRE-EMPTING

"High-energy bond" is biochemist's shorthand, not a physics statement. Energy is not stored inside the bond — the large negative ΔG of hydrolysis comes from relief of electrostatic repulsion between the negatively charged phosphates, greater resonance stabilization of the products, and better solvation of the products. Use the course's definition on the exam; just don't build a wrong mental picture on it.

1.5 Convergent versus divergent, and the summary table



Slide 12 — the shape of metabolism. (a) Starch, glycogen, sucrose, phospholipids, triacylglycerols and many amino acids all funnel into one node, **acetyl-CoA**. (b) From that same node diverge cholesterol (→ bile acids, steroid hormones, vitamin K), fatty acids → triacylglycerols, and phospholipids. (c) The TCA cycle regenerates its own oxaloacetate — a pathway that regenerates a component is a **cycle**, which is why cycles have no committed step (§3.2).

Feature	Catabolism	Anabolism
Shape	Convergent — many molecules → few end products	Divergent — few precursors → many products
End products	Simple molecules (CO ₂ , H ₂ O, NH ₃)	Complex molecules
What it does	Breaks something down	Builds something up
Energy	Releases energy, stored as ATP — exergonic	Requires energy, spends ATP/GTP — endergonic
Redox direction	Oxidative — NAD ⁺ /NADP ⁺ /FAD reduced to NADH/NADPH/FADH ₂	Reductive — NADPH oxidized, donating H to the growing molecule
Trigger	Stimulated by LOW energy charge	Stimulated by HIGH energy charge
Hormonal context	Carbohydrate burned at high insulin; fat at low insulin	Insulin — except gluconeogenesis and ketogenesis

SLIDE 14 — IN-CLASS QUESTION

"Anabolism and catabolism are 2 phases of metabolism. Anabolism:" A. refers to the breakdown of complex molecules to a few simple molecules · B. is generally exergonic · C. often involves oxidative reactions · **D. often requires NADPH** · E. produces NADH

Answer: D. Note the construction — **every distractor is a true statement about catabolism**, offered as if it described anabolism. This is the deck's favorite question format; watch for it.

Objective 2 / Key components of metabolism and the role of activated carriers

NUTRIENTS — WHY BOTH KINDS ARE NEEDED

Her framing: **you cannot have catabolism without nutrients, and you need both kinds**. Macronutrients are the fuel and the carbon skeletons. Micronutrients (vitamins and minerals) are **used by the enzymes** that metabolize the macronutrients — they enhance enzyme activity so those macronutrients can be efficiently utilized. Neither works without the other, which is why she always names them together.

STORED VS. MUST-BE-REPLENISHED (SLIDE 16)

STORED within the body	NOT stored — needed on a regular / near-daily basis
- Carbohydrate → as glycogen - Fat → as triacylglycerols - Amino acids → as muscle protein and the free amino acid pool Fat-soluble vitamins - Some minerals (e.g. iron)	Water-soluble vitamins This single fact explains the activated-carrier table below: most coenzymes come from water-soluble vitamins, which is why those vitamins must be eaten continually and why their deficiencies produce disease quickly.

FROM THE SPEAKER NOTES — NUTRITION AND DISEASE

Ill health can result from **dietary deficiency** (beriberi — thiamine/B₁; pellagra — niacin/B₃; protein-energy malnutrition) **or from excess** (e.g. alcohol). Some essential nutrients are **toxic, even deadly, in excess** — water, salt and iron. Most foods contain a mix of all nutrient types plus other compounds, including toxins.

WHAT A "METABOLIC PATHWAY" IS (SLIDE 17)

- A metabolic pathway **describes all the chemical transformations** of a given starting material.
- Reactions are **almost always catalyzed by enzymes** (protein molecules); very few reactions in organisms occur without catalysis.
- Reactants go **SUBSTRATE → PRODUCT**, and **the product of one enzyme is the substrate for the next enzyme**.
- Enzymes are **largely protein in nature** and convert substrate into product **per unit time** — this is velocity, the bridge into Lecture 2.

The slide illustrates this with glycolysis. Worth noticing on that figure: straight double-headed arrows are freely reversible steps run by one enzyme, while curved paired arrows mark steps whose forward and reverse reactions use **different** enzymes — those are the irreversible, regulated steps of Objective 3.

THE SIX RECURRING REACTION TYPES (SLIDE 18)

She walked this list verbally almost word for word.

Type of reaction	Description
Oxidation–reduction	Electron transfer
Ligation requiring ATP cleavage	Formation of covalent bonds (i.e. carbon–carbon bonds)
Isomerization	Rearrangement of atoms to form isomers
Group transfer	Transfer of a functional group from one molecule to another
Hydrolytic	Cleavage of bonds by the addition of water
Addition or removal of functional groups	Addition to double bonds, or removal to form double bonds

2.1 Activated carriers — the big memorization table

CONCEPT

Metabolism does not invent a new mechanism for every reaction. It uses a small set of reusable **shuttle molecules**, each carrying **one specific chemical group** and handing it off where needed. Learn the carrier and its cargo and you can predict what a reaction is doing the moment you see the carrier's name in it. The unifying fact — and the reason this sits in the same objective as the nutrition slide — is that **most of these carriers are coenzymes derived from water-soluble vitamins**. That is the entire link between "eat your B vitamins" and "run your metabolism." She called slide 19 "*a very important slide*" and read out every row.

Carrier molecule (activated form)	Group carried	Vitamin precursor
ATP	Phosphoryl	—
NADH and NADPH	Electrons	Nicotinate (niacin , B ₃)
FADH ₂	Electrons	Riboflavin (vitamin B ₂)
FMNH ₂	Electrons	Riboflavin (vitamin B ₂)
Coenzyme A	Acyl	Pantothenate (B ₅)
Lipoamide	Acyl	Lipoic acid
Thiamine pyrophosphate (TPP)	Aldehyde	Thiamine (vitamin B ₁)
Biotin	CO ₂	Biotin
Tetrahydrofolate (THF)	One-carbon units	Folate
S-Adenosylmethionine (SAM)	Methyl	—
Uridine diphosphate glucose (UDP-glucose)	Glucose	—
Cytidine diphosphate diacylglycerol (CDP-DAG)	Phosphatidate	—
Nucleoside triphosphates	Nucleotides	—

HOOKS TO MAKE THIS TABLE STICK

- **Niacin makes the N's** (NAD⁺, NADP⁺); **riboflavin makes the F's** (FAD, FMN). Two vitamins, four electron carriers.
- **Pantothenate is Part of CoA** — and CoA carries **acyl** groups, which is why acetyl-CoA is the universal carbon carrier.
- **Lipoamide** is the protein-bound form of lipoic acid and also carries acyl groups. CoA, lipoamide, TPP, FAD and NAD⁺ all appear together in one enzyme complex — pyruvate dehydrogenase — later in the course.
- **Biotin = CO₂**. Whenever you see a **carboxylase**, think biotin.
- **THF = one-carbon units** generally; **SAM = methyl** specifically. Her line: "*Sam is not your next-door neighbor — SAM is the active carrier of methyl groups.*"
- **UDP-glucose** carries glucose and is used for **glycogen synthesis** — she named this application explicitly.
- The rows with **no vitamin precursor** are the nucleotide-derived ones: ATP, SAM, UDP-glucose, CDP-DAG, nucleoside triphosphates.

THREE NOTES ON THIS SLIDE

- She pronounced SAM as "*acetyl-S-methionine*." It is **S-adenosyl-methionine**, and it donates a **methyl**, not an acetyl, group — acetyl transfer is coenzyme A's job.
- "Biotin is used as such" is *not* a slip: biotin is one of the rare cases where the vitamin and the coenzyme have the same name and essentially the same structure.
- The leftover title text box on slide 19 reads "Lipoic Acid" — ignore it; the slide is the full carrier table.

SLIDE 20 — HER FOUR-LINE SUMMARY OF OBJECTIVE 2

1. Nutrients are linked to biochemical pathways.
2. Biochemical pathways are organized into metabolic pathways.
3. Key reactions repeated throughout the metabolic pathways are catalyzed by enzymes.
4. Many of the activated carriers in metabolism are derived from **water-soluble vitamins**.

Objective 3 / Common themes; committed vs. rate-limiting steps

THE FIVE COMMON THEMES (SLIDE 22)

Every metabolic pathway you will meet is a **multi-step, enzyme-catalyzed sequence**: $A \xrightarrow{E_1} B \xrightarrow{E_2} C \xrightarrow{E_3} D \xrightarrow{E_4} E$, in which some arrows are double-headed (reversible) and some single-headed (irreversible). Five themes recur in all of them: ① **irreversibility** ② **committed step** ③ **rate-limiting step** ④ **regulation** ⑤ **compartmentalization**.

3.1 Irreversibility — the theme everything hangs on

SLIDE 23

If the reaction sequences for anabolic and catabolic pathways were identical: ① their **simultaneous operation would be wasteful**, and ② they **could not be separately regulated**. Therefore metabolic pathways have **one or more steps that are essentially irreversible under physiological conditions**.

HER WAY OF EXPLAINING IT — THE BOTTLENECK AND THE FUTILE CYCLE

An irreversible step is the **bottleneck of the pathway**: block it and the entire pathway slows. That is why irreversible steps are the ones that get regulated — they are the only place where a single intervention controls the whole flow.

Her worked example, used four times: **fatty acid synthesis and fatty acid oxidation**. You do not want both running at once — the cell would build fat and immediately burn it, spending ATP for nothing, and the purpose of fat metabolism would become **futile**. So when synthesis is on, oxidation slows, and vice versa. Irreversible, separately regulated steps are what make that switching possible.

Two qualifiers she stated twice: **a pathway can have more than one irreversible step**, and **steps other than the committed and rate-limiting ones can also be regulated**.

"None of the pathways in your body is shut down. One is active over the other — one is active and the other is slowed down."

TWO DIFFERENT PROPERTIES SHE BLENDED TOGETHER

Describing the rate-limiting step, she said it "has a very high activation energy ... they have a very high delta G and is negative and that is why they are irreversible." Those are **two separate physical properties**, and exams ask about them separately:

Property	Branch	What it determines
ΔG — large and negative	Thermodynamics	Direction . The reaction goes one way and effectively does not come back → the step is irreversible .
E_a — highest in the pathway	Kinetics	Speed . A high energy barrier makes the step the slowest → the step is rate-limiting .

In practice these usually land on the same step, which is why her sentence works — but "why is it irreversible?" and "why is it slow?" have different answers. You formalize ΔG in the Thermodynamics primer this same week.

3.2 Committed step

SLIDE 24, ALL FOUR BULLETS

- Early in each linear pathway there is usually an **irreversible** reaction that **commits the product to continue** down that pathway.
- It is **generally the target of regulation**.
- In many instances, **but not always**, the committed step is **also** the rate-limiting step.
- **Circular pathways do not have a committed step** — though they *do* have a rate-limiting step.

WHY A CYCLE CAN'T HAVE A COMMITTED STEP

"Committed" means that once a molecule crosses this step it has **only one possible fate**. In a cyclic pathway the intermediates are **regenerated** — oxaloacetate becomes citrate becomes ... becomes oxaloacetate again — so nothing is ever irreversibly committed to a single destination. A cycle still has a slowest step, hence a rate-limiting step, but no point of no return.

3.3 Rate-limiting step

SLIDE 25, ALL FIVE BULLETS

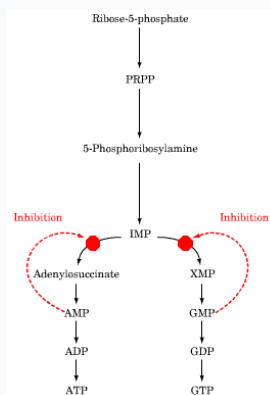
- The **slowest reaction** in a metabolic pathway.
- **Generally has the highest activation energy** of all reactions in the pathway.
- Its **velocity determines the overall flux of metabolites** through the pathway.
- **Usually subject to intense regulation, both positive and negative**.
- **Rate-limiting steps are regulated, but a regulated step may not be rate-limiting**.

The enzyme catalyzing it is the **rate-limiting enzyme** — a term you will use constantly for the rest of the course.

THE LOGIC RULE SHE STATED FOUR SEPARATE TIMES

rate-limiting $\Rightarrow$ regulated **TRUE** | regulated $\Rightarrow$ rate-limiting **FALSE**

In her words: "rate limiting steps are always regulated, subjected to intense regulation — but the regulated steps may not be rate limiting always." The counterexample is built into the slide, below.



Slide 25's inset — the pathway that proves the rule. Purine synthesis: ribose-5-P $\rightarrow$ [regulated] $\rightarrow$ PRPP $\rightarrow$ [rate-limiting AND regulated] $\rightarrow$ 5-phosphoribosylamine $\rightarrow$ IMP $\rightarrow$ AMP and GMP, both feeding back to inhibit. **Step 1 is regulated but not rate-limiting; step 2 is both.**

	Committed step	Rate-limiting step
Reversibility	Irreversible	Irreversible
Position	Early in a linear pathway	Anywhere in the pathway
Defining property	Commits the metabolite to that pathway — no turning back, no diversion	The slowest step; highest activation energy
What it controls	Whether material enters the pathway at all	The overall flux through the pathway
Regulation	Generally the target of regulation	Intense regulation, positive and negative
In cyclic pathways	Does not exist	Does exist
Relationship	Often the same step — but not always	

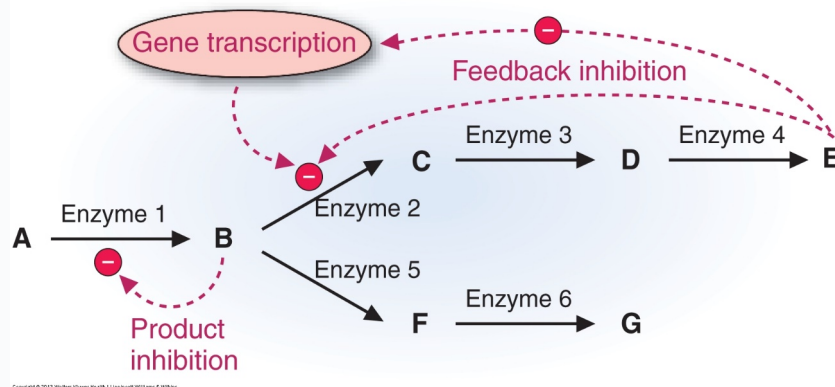
SLIDE 26 — HER SUMMARY OF OBJECTIVE 3

- Certain enzymes catalyze irreversible reactions, and those enzymes can be influential in regulation.
- **Committed step** — irreversible, generally the target of regulation, early in a linear pathway.
- **Rate-limiting step** — irreversible, subject to intense regulation, slowest in the metabolic pathway.

Objective 4 / Mechanisms of regulation of metabolic pathways

THE FRAME

Enzyme activity can be regulated by **activation or inhibition**, to **control metabolic pathways to match the body's requirements** moment by moment. Everything here applies primarily to the enzymes catalyzing the **irreversible** (committed / rate-limiting) steps — though, as she repeated, other steps can be regulated too.



Slide 28 — three mechanisms in one picture. **Product inhibition:** the immediate product B inhibits enzyme 1, which just made it. **Feedback inhibition:** the end product E inhibits enzyme 2, the first enzyme *after the branch point*, shutting down only its own branch while F→G keeps running. **Gene transcription:** E also represses the enzyme's gene — the slow, long-term arm.

TWO TERMS STUDENTS CONSTANTLY SWAP

Term	Inhibition by...
Product inhibition	the IMMEDIATE product — the very next molecule made by that enzyme
Feedback inhibition	the FINAL / end product of the pathway, acting anywhere upstream at the level of an enzyme

Her plain-English gloss on product inhibition: "*the product is saying — enough has been formed, slow down.*"

4.1 The master split: long-term vs. short-term

LONG term regulation alters enzyme QUANTITY · SHORT term regulation modifies enzyme ACTIVITY

	Long-term regulation	Short-term regulation
What changes	The amount of enzyme protein present	The activity of enzyme molecules that already exist
Where it acts	At the level of the gene (transcription) and protein degradation	On the existing enzyme protein
Speed	Hours to days	Minutes to hours
Reversible?	Only by re-synthesis / re-degradation	Reversible and rapid — one exception (proenzyme activation)
Mechanisms	Induction · Repression · Degradation	Substrate concentration · Product inhibition · Proenzyme activation · Reversible covalent modification · Allosteric regulation
Role	Sets the cell's long-run capacity for a pathway	Carries out most of the moment-to-moment physiological regulation

LONG-TERM REGULATION — THE SPECIFICS (SLIDES 29–30)

- Regulation of gene expression controls the **quantity and rate of enzyme synthesis**; a slow process taking hours to days.
- **INDUCTION** — increasing the rate of enzyme synthesis by **enhancing the rate of gene transcription** → more enzyme.
- **REPRESSION** — decreasing the rate of enzyme synthesis by **decreasing the rate of gene transcription** → less enzyme.
- **DEGRADATION** — by the **ubiquitin/proteasome pathway** and the **lysosomal pathway**. These are the body's only two major routes of protein degradation; memorize the pair.
- **Insulin is an anabolic hormone: it INDUCES the key enzymes of anabolic pathways.**
- Speaker note: this applies mainly to **enzymes used once at a particular stage**, rather than enzymes in constant use.

SHORT-TERM REGULATION — THE FIVE MECHANISMS (SLIDE 31, IN THE DECK'S ORDER)

1. Effect of **substrate concentration**
2. **Product inhibition**
3. **Activation of pre-existing pools of inactive pro-enzymes** to produce active enzymes
4. **Reversible covalent modification**
5. **Allosteric regulation**

Short-term regulation **usually does not affect the concentration of enzyme protein**; it is **reversible and rapid**, and carries out most moment-to-moment regulation of enzyme activity.

THE EXCEPTION, FLAGGED IN HER OWN SPEAKER NOTES

Proenzyme activation is the odd one out. It is short-term and rapid, but because it activates pre-formed proenzymes it **does increase the concentration of active enzyme** and is **NOT reversible**. Every other short-term mechanism is reversible and leaves enzyme concentration alone.

Mechanisms 1 & 2 — Substrate concentration (glucokinase) and product inhibition (hexokinase)

SLIDES 32–33

- The velocity of ALL enzymes is dependent on the concentration of substrate.
- Liver glucokinase phosphorylates large amounts of glucose for **glycogen synthesis**. **Fasting**: [glucose] low → glucokinase inactive. **After a meal**: [glucose] rises → glucokinase activated.
- Peripheral tissue hexokinase: during **fasting**, [glucose] is low **but** hexokinase still efficiently phosphorylates glucose. If **glucose-6-phosphate accumulates**, it **inhibits** hexokinase.
- NB — in the liver, glucokinase is NOT inhibited by glucose-6-phosphate.

WHY THIS WORKS — BUILD THE STORY THE WAY SHE DID

The liver is the chief site of metabolism. After a meal, glucose is absorbed and reaches the liver via transporters. Once inside it must be committed, and the first step of essentially any glucose metabolism is **adding a phosphate** — the enzymes that add phosphate are **kinases**. Phosphorylating glucose to **glucose-6-phosphate** does two things at once: it puts a negative charge on the molecule so it **cannot leave the cell** (her phrase: it prevents "irreversible exit," meaning the glucose is trapped inside), and it activates it for the downstream pathways.

The underlying mechanism, formalized in Lecture 2, is K_m . Glucokinase has a **high K_m** (~10 mM), well above normal blood glucose, so it only works meaningfully when portal glucose is high — which is *why* substrate concentration is its regulator. Hexokinase has a **low K_m** (~0.1 mM) and is essentially saturated at normal glucose, so substrate concentration tells it nothing and it needs a different control — its product.

THE PAIRED-ENZYME GRID — A NEAR-CERTAIN EXAM ITEM

She spent several minutes insisting that **not all enzymes undergo the same kind of regulation**, and used this pair as the proof. Know the grid in both directions.

	Hexokinase	Glucokinase
Tissue	Most / peripheral tissues	Liver (and pancreatic β cells)
K_m for glucose	Low (~0.1 mM) — high affinity	High (~10 mM) — low affinity
Active when?	Essentially always — works even in fasting	Only when glucose is high (after a meal)
Regulated by substrate concentration?	NO — already saturated	YES — this is its regulation
Inhibited by glucose-6-P?	YES — classic product inhibition	NO
Induced by insulin?	No	Yes — so it also appears under long-term regulation

ONE WORD OF PRECISION

"Glucokinase is inactive during fasting" is the answer the course wants. Strictly, the enzyme is present and functional but, because of its high K_m , operates far below V_{max} at fasting glucose — so its contribution is negligible. Same practical conclusion, cleaner mechanism. (It is also sequestered in the nucleus by glucokinase regulatory protein when glucose is low, but that is beyond this lecture.)

Mechanism 3 — Pro-enzyme (zymogen) activation

SLIDE 34

- Pro-enzymes are also known as **zymogens**.
- Activation is a **rapid** way to increase enzyme activity, but has the **disadvantage of not being a reversible process**.
- Canonical example: **pancreatic proteases**. Pro-enzymes are **synthesized in abundance, stored in secretory granules, and covalently activated on release** at the site of action.
- The cascade: **trypsinogen** → **trypsin** (releasing **TAP**, trypsinogen activation peptide), and trypsin then **amplifies** by activating every other pancreatic zymogen — chymotrypsinogen→chymotrypsin, proelastase→elastase, kallikreinogen→kallikrein, procarboxypeptidase→carboxypeptidase, prophospholipase→phospholipase, procolipase→colipase.
- Activation routes on the figure: **normal** — **enterokinase (enteropeptidase)** in the **brush border of the small intestine**, plus trypsinogen **autoactivation** (a unique feature of human trypsinogen); **abnormal** — **cathepsin B, located within acinar cells**.
- Speaker note: the **blood clotting cascade** is another example — each factor activated by proteolysis catalyzed by the previous factor.

CLINICAL CORRELATION — ACUTE PANCREATITIS (SLIDE 35)

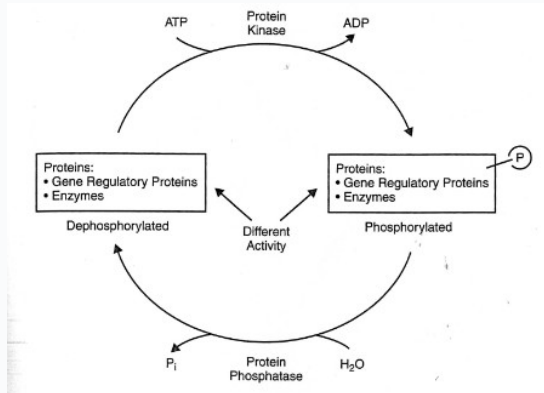
The logic: pancreatic proteases are proteolytic. Active inside the acinar cell, they would digest the cell that made them. So the pancreas builds them as inactive zymogens, stores them, and activates them only in the **lumen of the intestine**, at the site of action.

When that fails: inappropriate **intracellular** activation of trypsinogen → active trypsin inside the acinar cell → autoactivation of the whole protease cascade → **pancreatic cell autolysis** → **acute pancreatitis**. She noted this can also be an **inherited** condition. Her stated philosophy, repeated: "*we are not learning pure biochemistry — we are learning biochemistry applicable to clinical medicine.*"

Mechanism 4 — Reversible covalent modification

SLIDES 36–37

- **Rapid and transient** regulation of enzyme activity, by addition or removal of a phosphate.
- **Protein kinase** adds the phosphate ($\text{ATP} \rightarrow \text{ADP}$); **phosphoprotein phosphatase** removes it ($\text{H}_2\text{O} \rightarrow \text{releases } \text{P}_i$).
- **Usual sites: the hydroxyl group of SERINE, THREONINE and TYROSINE residues** of the key enzyme. The phosphate lands on an $-\text{OH}$, which is exactly why those three residues.
- It switches **entire sets of opposing metabolic pathways** on or off at once. Depending on where you are in the **fed–fasting cycle: ALL** target proteins are phosphorylated by protein kinases, **OR ALL** are dephosphorylated by phosphoprotein phosphatases.
- The **phosphorylated form may be active or inactive**, and likewise the dephosphorylated form. **It depends on the specific enzyme, and you must learn each one as it comes.**
- Speaker note — other groups that can be attached covalently: **phosphoryl, methyl, uridylyl, adenylyl, ADP-ribosyl**.



Slide 37 — who pushes which button. Glucagon and epinephrine drive the top arrow (protein kinase → phosphorylated); insulin drives the bottom arrow (phosphatase → dephosphorylated). The targets are both **enzymes and gene regulatory proteins**, and the two forms have different activity.

INSULIN always Dephosphorylates · GLUCAGON and EPINEPHRINE always PHOSPHORYLATE

THE SINGLE HIGHEST-YIELD SENTENCE IN OBJECTIVE 4

Her words: "Insulin always and always dephosphorylates the key enzyme. Please don't mess up there. Insulin never phosphorylates the key enzymes. Glucagon and epinephrine always and always phosphorylate the key enzyme."

The corollary she derived, which is the useful form: since insulin always dephosphorylates, **any pathway supported by insulin is active in the DEPHOSPHORYLATED form** — and any pathway driven by glucagon/epinephrine is active in the phosphorylated form. You can now predict the phosphorylation state of a regulated enzyme from nothing but knowing which hormone turns its pathway on.

TWO CLARIFICATIONS ON SLIDE 37

- The slide reads "*the counter-regulatory hormones (glucagon and insulin).*" That is loose: **insulin is not a counter-regulatory hormone** — glucagon, epinephrine, cortisol and growth hormone are counter-regulatory *to insulin*. The intended meaning, which she stated correctly out loud, is that insulin and its opponents together control most of intermediary metabolism.
- **Why** they act in opposite directions (Lippincott background, not required here): glucagon and epinephrine bind **G protein-coupled receptors** → G_s → **adenyl cyclase** → **cAMP** → **protein kinase A** → phosphorylates Ser/Thr. Insulin acts through a receptor tyrosine kinase and ultimately activates **phosphoprotein phosphatases**. Covered fully in Signal Transduction later this same week.

Mechanism 5 — Allosteric regulation

WHAT SHE SAID ABOUT IT HERE

- **Allosteric modifiers** alter enzyme activity by binding at a site **other than the active site**.
- Two kinds: **allosteric activators** (increase activity) and **allosteric inhibitors** (decrease activity). Time frame: **minutes**.
- She deferred the mechanism explicitly — "*which will be taught with enzymes*", i.e. Lecture 2, in the very same session.

4.2 Compartmentalization

CONCEPT

Physically separating two opposing pathways is itself a form of regulation. If fatty acid synthesis and fatty acid oxidation happen in different places, they can have different enzymes, different carriers and different controls — and cannot short-circuit each other into a futile cycle. It operates at two levels, and the second has a diagnostic payoff.

SLIDE 38 — BOTH LEVELS WITH THEIR EXAMPLES

Level	Purpose	Examples she gave
Cellular	To segregate catabolic and anabolic pathways in the same cell . Pathways are cytosolic / mitochondrial / both .	Fatty acid OXIDATION → mitochondria Fatty acid SYNTHESIS → cytoplasm
Tissue	To carry out organ-specific metabolic processes . Also a clinical indicator of tissue damage .	Urea synthesis → liver Steroid hormone synthesis → adrenals

CLINICAL CORRELATION — USING COMPARTMENTALIZATION TO LOCALIZE DAMAGE

Her worked example: the urea cycle operates in the **liver**, and urea is reported clinically as **blood urea nitrogen (BUN)**. A damaged liver cannot make urea — so a **low BUN** can point to **liver damage** (alcohol, toxins, any liver insult).

Round this out so you don't get caught: low BUN's causes are advanced liver failure, low protein intake/malnutrition, and overhydration. A **high BUN** is the far more common clinical finding and points elsewhere — kidney impairment, dehydration, or an upper GI bleed. She is making a biochemical point (no urea cycle → no urea), not a full differential.

The general principle, which you will use all year: because pathways are compartmentalized to specific tissues, finding a tissue-specific enzyme in the blood tells you *which* tissue was damaged. This is the entire basis of the enzyme panels in clinical medicine.

4.3 Timeline of regulation

Slide 39 — reproduce this exactly; it is the most compressed, most testable content in the lecture.

Event	Time
Substrate stimulation	Minutes
Product inhibition	Minutes
Allosteric regulation	Minutes
Covalent modification	Minutes – Hours
Synthesis or degradation	Hours – Days

THE ONE RULE THAT GENERATES THE WHOLE TABLE

If it touches the gene, it takes hours to days. If it touches an enzyme that already exists, it takes minutes.

Covalent modification straddles the boundary because a phosphorylation cascade can be near-instant but hormone-driven whole-body switching takes longer to play out. Only the last row is long-term regulation; the top four are all short-term.

SLIDE 41 – THE CLOSING IN-CLASS QUESTION

"Hormones regulate certain metabolic pathways by altering the rate of expression of genes that encode enzymes. What would be the time frame of such regulation?"

Answer: Regulation of gene expression alters the **amount** of the protein encoded by that gene, so the time frame is **several hours to days** — this is **long-term regulation**, acting by **induction or repression**, changing enzyme **quantity** rather than modifying activity.

SLIDE 40 – HER SUMMARY OF OBJECTIVE 4

- General mechanisms regulate enzyme-catalyzed reactions, each with its own time course; regulation is **long-term** or **short-term**.
- Short-term regulation is usually **reversible and rapid**.
- **Proenzyme activation is not reversible**; inappropriate activation causes **pancreatitis**.
- **Compartmentation** of enzymes into multienzyme complexes or organelles is itself a means of regulation.

Part 5 / Where the lecture was imprecise — reconciled

Nothing here changes an exam answer. It is here so that when you meet the correct version in the textbook or on a board-style question, you recognize it rather than assuming you learned it wrong.

What was said / shown	What is precisely true	Does it matter for the exam?
SAM described as "acetyl-S-methionine"	S-adenosyl-methionine ; it donates a methyl group	Yes — know the name and that its cargo is methyl
Rate-limiting step "has high activation energy ... high negative ΔG and that is why it is irreversible"	Two different properties: large negative ΔG → irreversible (thermodynamics); highest E_a → slowest / rate-limiting (kinetics)	Yes if asked "why irreversible" vs "why slow." Usually the same step, so her sentence works in practice
Slide 37: "the counter-regulatory hormones (glucagon and insulin)"	Insulin is not counter-regulatory . Glucagon, epinephrine, cortisol and GH are counter-regulatory to insulin	Only if a question uses the term precisely — but worth knowing
"Glucokinase is inactive in fasting"	Present and functional, but its high K_m means it operates far below V_{max} at fasting glucose, so its contribution is negligible	No — answer "inactive"; the K_m reasoning is the <i>why</i>
"NADH is the source of ATP"	NADH is oxidized in the ETC; the proton gradient it builds drives ATP synthase. She said " indirect source," which is exactly right	No — her wording is already correct
"Reducing equivalents are hydrogen, $2H^+$ and 2 electrons"	That is what is removed from the substrate . NAD^+ then takes a hydride and a free H^+ is released → $NADH + H^+$; FAD takes both hydrogens → $FADH_2$	Only for why NADH is written " $NADH + H^+$ " — which she was careful about
"Energy stored in the high-energy bond"	The large negative ΔG of hydrolysis comes from charge-repulsion relief, resonance stabilization and solvation of products	No — use the course definition; just don't build a wrong mental model
HMP shunt grouped as one of the three "oxidative pathways"	Only its first (oxidative) phase is oxidative and irreversible; the non-oxidative phase is reversible sugar interconversion. "Alternate pathway of oxidation" is fair	No — the triad glycolysis / TCA / HMP shunt is what she is testing

Part 6 / Self-test

Cover the answers. Say each one out loud before you look — the goal is recall, not recognition.

- 1 Name the three oxidative pathways, what each handles, and which yields no ATP. **Glycolysis (mainly carbohydrate); TCA (common to carbohydrate, lipid, protein, from acetyl-CoA); HMP shunt — NADPH + pentoses, NO ATP**
- 2 Oxidation in the body occurs by what change, and what are the three acceptors? **Loss of hydrogen. NAD^+ , NADP^+ , $\text{FAD} \rightarrow \text{NADH} + \text{H}^+$, NADPH , FADH_2 .**
- 3 List the four requirements for anabolism. **ATP (universal; GTP for some), acetyl-CoA (carbon), NADPH (hydrogen), insulin.**
- 4 Which two "-geneses" does insulin not support, and why? **Gluconeogenesis and ketogenesis — both are fasting/low-insulin programs, so insulin inhibits them.**
- 5 How many high-energy bonds in ATP and ADP, and what is ΔG° of hydrolysis? **Two (β and γ phosphoanhydride) and one. -7.3 kcal/mol for both. The α -to-ribose bond is a phosphoester and is not high energy.**
- 6 Energy charge: range, real-cell range, and effect of a high value? **0 (all AMP) to 1 (all ATP); buffered at 0.8–0.95. High EC inhibits catabolic and stimulates anabolic pathways.**
- 7 Which nutrients are stored, and which must be consumed regularly? **Stored: glycogen, triacylglycerol, muscle protein + free amino acid pool, fat-soluble vitamins, some minerals (iron). Not stored: water-soluble vitamins.**
- 8 Vitamin precursor for $\text{NAD}^+/\text{NADP}^+$, FAD/FMN, CoA, TPP, THF, lipoamide? **Niacin; riboflavin; pantothenate; thiamine; folate; lipoic acid.**
- 9 Group carried by biotin, THF, SAM, UDP-glucose, CoA, ATP? **CO_2 ; one-carbon units; methyl; glucose; acyl; phosphoryl.**
- 10 Define the committed step — where is it found and where never? **Irreversible reaction early in a linear pathway committing the product to that pathway; generally the target of regulation; often but not always also rate-limiting. Never in circular pathways, which still have a rate-limiting step.**
- 11 State the logical relationship between "rate-limiting" and "regulated." **Rate-limiting $\Rightarrow$ regulated (always). Regulated $\Rightarrow$ rate-limiting (NOT always).**
- 12 Why can't anabolic and catabolic pathways use identical reaction sequences? **Simultaneous operation would be wasteful (futile cycling), and they could not be separately regulated.**
- 13 Contrast glucokinase and hexokinase in four ways. **GK: liver, high K_m , substrate-regulated (active only after a meal), NOT inhibited by G6P, insulin-induced. HK: peripheral, low K_m , works in fasting, IS inhibited by G6P, not substrate-regulated.**
- 14 Product inhibition vs feedback inhibition? **Product = by the immediate product of that enzyme. Feedback = by the final product of the pathway.**
- 15 Which short-term mechanism is irreversible, and what happens when it misfires? **Proenzyme (zymogen) activation. Intracellular trypsinogen activation $\rightarrow$ trypsin $\rightarrow$ acinar cell autolysis $\rightarrow$ acute pancreatitis.**
- 16 Which residues are phosphorylated, by which enzyme, and what reverses it? **The hydroxyls of serine, threonine, tyrosine. Protein kinase adds (ATP); phosphoprotein phosphatase removes (H_2O , releasing P_i).**
- 17 What do insulin, glucagon and epinephrine do to key enzymes? **Insulin always dephosphorylates; glucagon and epinephrine always phosphorylate. So insulin-supported pathways are active dephosphorylated.**
- 18 Name the long-term mechanisms and the two protein degradation routes. **Induction and repression ($\pm$ gene transcription $\rightarrow \pm$ enzyme quantity). Degradation via ubiquitin/proteasome and lysosomal pathways.**

- 19 Give one cellular and one tissue example of compartmentalization, plus one clinical use. **Fatty acid oxidation (mitochondria) vs synthesis (cytoplasm); urea synthesis in liver, steroid hormones in adrenals; low BUN can indicate liver damage.**
- 20 Time course for substrate stimulation, product inhibition, allosteric regulation, covalent modification, synthesis/degradation? **Minutes; minutes; minutes; minutes–hours; hours–days.**

Part 7 / Rapid-review sheet

Everything in **bold** is one of Dr. Bagh's "always / never" statements — the ones she repeated most, or flagged with "please know that," "kindly remember," "don't mess up there," "very very important." If you review one page before the exam, review this one.

Catabolism vs anabolism

Catabolism: convergent · complex→simple · exergonic · oxidative · releases energy → ATP · **triggered by LOW energy charge**

Anabolism: divergent · simple→complex · endergonic · reductive · spends ATP · **triggered by HIGH energy charge** — **never a low energy state** · insulin

Oxidation in the body

= **LOSS OF HYDROGEN, not gain of oxygen** (dehydrogenation)

$\text{NAD}^+ \rightarrow \text{NADH} + \text{H}^+$ · $\text{NADP}^+ \rightarrow \text{NADPH}$ · $\text{FAD} \rightarrow \text{FADH}_2$

NADH and FADH₂ → ATP (indirectly). **NADPH → NEVER ATP** — it is the H donor for reductive synthesis

The three oxidative pathways

Glycolysis — mainly carbohydrate → acetyl-CoA

TCA cycle — **common to carbohydrate + lipid + protein**; starts at acetyl-CoA → $\text{CO}_2 + \text{H}_2\text{O}$, NADH, FADH_2

HMP shunt — alternate; **NADPH + pentoses, NO ATP**

Anabolism needs

① **ATP** (universal; sometimes GTP) ② **acetyl-CoA** = carbon ③

NADPH = hydrogen ④ **↑ insulin**

Insulin supports every "-genesis" EXCEPT gluconeogenesis and ketogenesis

ATP / energy charge

ATP = **2** high-energy (phosphoanhydride) bonds; ADP = **1**; $\Delta G^{\circ'} = -7.3 \text{ kcal/mol}$

EC = $([\text{ATP}] + \frac{1}{2}[\text{ADP}]) / ([\text{ATP}] + [\text{ADP}] + [\text{AMP}])$; range **0–1**; buffered at **0.8–0.95**

High EC → catabolism ↓, anabolism ↑

Activated carriers

ATP → phosphoryl

NADH/NADPH → electrons — **niacin**

FADH_2 , FMNH_2 → electrons — **riboflavin**

Coenzyme A → acyl — **pantothenate**

Lipoamide → acyl — **lipoic acid**

TPP → aldehyde — **thiamine**

Biotin → CO_2 — **biotin**

THF → one-carbon — **folate**

SAM → methyl · UDP-glucose → glucose

CDP-DAG → phosphatidate · NTPs → nucleotides

Most are coenzymes from water-soluble vitamins.

Five common themes

Irreversibility · committed step · rate-limiting step · regulation · compartmentalization

A pathway may have more than one irreversible step, and steps other than the committed/rate-limiting ones can also be regulated.

Committed vs rate-limiting

Committed: irreversible · early · linear pathways only · target of regulation · **absent from cycles**

Rate-limiting: irreversible · **slowest** · **highest E_a** · sets overall flux · intensely regulated

Rate-limiting ⇒ regulated. Regulated ⇏ rate-limiting.

No pathway is ever fully shut down — one is active while the opposing one slows.

Long vs short term

LONG = enzyme QUANTITY · gene level · **hours–days** · induction / repression / degradation (ubiquitin-proteasome, lysosomal)

SHORT = enzyme ACTIVITY · **minutes–hours** · reversible & rapid

Five short-term mechanisms

① substrate concentration — **glucokinase**

② product inhibition — **hexokinase ← G6P**

③ proenzyme activation — **NOT reversible**

④ reversible covalent modification — Ser/Thr/Tyr –OH

⑤ allosteric regulation — activators / inhibitors

Product inhibition = immediate product. Feedback inhibition = final product.

Hexokinase vs glucokinase

HK: peripheral · low K_m · always on · **inhibited by G6P**

GK: liver · high K_m · **only after a meal** · **NOT inhibited by G6P** · induced by insulin

Phosphorylation rule

Insulin → always **DEphosphorylates** (phosphatase)

Glucagon / epinephrine → always **PHOSPHORYLATE** (kinase, uses ATP)

Sites: **Ser, Thr, Tyr –OH**. At any moment ALL targets are phosphorylated OR ALL dephosphorylated.

Insulin-driven pathways are active dephosphorylated.

Compartmentalization

Cellular: FA oxidation = **mitochondria**; FA synthesis = **cytoplasm**

Tissue: urea = **liver**; steroid hormones = **adrenals**

Clinical: low BUN → possible liver damage

Timeline

Substrate stimulation — minutes

Product inhibition — minutes

Allosteric regulation — minutes

Covalent modification — minutes–hours

Synthesis / degradation — hours–days

Touches the gene → hours–days. Touches an existing enzyme → minutes.

Clinical hooks

Acute pancreatitis — intracellular trypsinogen activation → acinar autolysis (can be inherited; cathepsin B route)

Low BUN — liver damage (urea cycle is hepatic)

Beriberi — thiamine · **Pellagra** — niacin · **PEM** — protein-energy malnutrition

Toxic in excess: alcohol, water, salt, iron

SOURCES RECONCILED

Lecture 01_Bagh_Overview of Metabolic Pathway.pptx — all 41 slides plus every speaker note. **Lecture 1 transcript Biochem.txt** — the full spoken lecture. **DO Med Biochem Genetics I syllabus, Fall 2026 (7/22)** — scheduling, readings, exam scope. **Lippincott Illustrated Reviews: Biochemistry, Ch. 8** — the required chapter, used for the three stages of catabolism and the hormone-signaling background. Figures reproduced from the lecture slides; original credits shown there are Berg/Tymoczko/Stryer *Biochemistry* 5th ed., Lippincott Williams & Wilkins, Lieberman Marks' *Basic Medical Biochemistry* 4th ed., and Frossard & Pastor, *Frontiers in Bioscience* 7:d275–287 (2002).